Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The degree of the differential equation $\frac{dy}{dx} - x = \left(y - x\frac{dy}{dx}\right)^{-4}$ is

(A) 2

(B) 3

(C) 4

(D) 5

Q.2

The degree of differential equation satisfying the relation
$\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda\left(x\sqrt{1+y^2} - y\sqrt{1+x^2}\right)$
is

(A) 1

(B) 2

(C) 3

(D) 4

Q.3

What is the order of the differential equation
$\frac{dx}{dy} + \int ydx = x^3$?

(A) 1

(B) 2

(C) 3

(D) Cannot be determined

Q.4

The order and degree of the differential equation of parabolas having vertex at the origin and focus at $(a, 0)$ where $a > 0$, are respectively

(A) 1,1

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(B) 2,1

(C) 1,2

(D) 2,2

Q.5

The degree and order respectively of the differential equation $\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{1}{\mathrm{x+y+1}}$ are

(A) 1,1

(B) $1, 2$

(C) 2,1

(D) 2,2

Q.6

What is the degree of the differential equation $y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^{-1}$?

(A) 1

(B) 2

(C) -1

(D) Degree does not exist.

Q.7

The differential equation $\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \sin y + x^2 = 0$ is of the following type

(A) Linear

(B) Homogeneous

(C) Order two

(D) Degree two

Q.8

What is the degree of the differential equation $\mathrm{k}\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2} = \left[1 + \left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)^3\right]^{3/2}$, where k is a constant?

(A) 1

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(B) 2

(C) 3

(D) 4

Q.9

What is the degree of the differential equation
$\left(\frac{d^3y}{dx^3}\right)^{2/3} + 4 - 3\left(\frac{d^2y}{dx^2}\right) + 5\left(\frac{dy}{dx}\right) = 0$?

(A) 3

(B) 2

(C) $2/3$

(D) Not defined

Q.10

What are the order and degree respectively of
the differential equation $\mathrm{y} = \mathrm{x}\frac{\mathrm{dy}}{\mathrm{dx}} + \frac{\mathrm{dx}}{\mathrm{dy}}$?

(A) 1,1

(B) 1,2

(C) 2,1

(D) 2,2

Q.11

Consider a differential equation of order m and degree n. Which one of the following pairs is not feasible ?

(A) $(3, 2)$

(B) $(2, 3/2)$

(C) $(2, 4)$

(D) $(2, 2)$

Q.12

Consider the following statements in respect of the differential equation
$\frac{d^2y}{dx^2} + \cos\left(\frac{dy}{dx}\right) = 0$
Statement 1: The degree of the differential equation is not defined.
Statement 2: The order of the differential equation is 2 .
Which of the above statements is/are correct?

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(A) 1 only

(B) 2 only

(C) Both 1 and 2

(D) Neither 1 nor 2

Q.13

The differential equation representing the family of curves $y^2 = 2c(x + c^{\frac{1}{2}})$, where $C > 0$, is a parameter, is of order and degree as follow

(A) order 1 , degree 3

(B) order 2, degree 2

(C) order 1, degree 2

(D) order 1 , degree 1

Q.14

The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants is of

(A) second order and first degree

(B) second order and second degree

(C) first order and second degree

(D) first order and first degree

Q.15

The degree and order of the differential equation of the family of all parabolas whose axis is x -axis, are respectively

(A) 2,1

(B) $1, 2$

(C) 3,2

(D) $2, 3$

Answers & Solutions

Q.1 Answer:

5

Solution:

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$\because \frac{dy}{dx} - x = \left(y - x\frac{dy}{dx}\right)^{-4}$

$\Rightarrow \left(\frac{\mathrm{dy}}{\mathrm{dx}} - \mathrm{x}\right)\left(\mathrm{y} - \mathrm{x} \cdot \frac{\mathrm{dy}}{\mathrm{dx}}\right)^{4} = 1$

- Order of the above differential equation $= 1$

& **degree** $= 5$

Q.2 Answer:

1

Solution:

$\sqrt{1+x^2} + \sqrt{1+y^2} = \lambda\left(x\sqrt{1+y^2} - y\sqrt{1+x^2}\right)$

$\Rightarrow \sqrt{1+x^2}(1+\lambda y) = \sqrt{1+y^2}(\lambda x - 1)$

$\Rightarrow \frac{\sqrt{1+x^2}}{\sqrt{1+y^2}} = \frac{\lambda x-1}{\lambda y+1} \Rightarrow \frac{x^2+1}{y^2+1} = \frac{\lambda^2x^2-2\lambda x+1}{\lambda^2y^2+2\lambda y+1}$

$\Rightarrow \left(y^2+1\right)\left(\lambda^2x^2 - 2\lambda x + 1\right)$

$= \left(x^2+1\right)\left(\lambda^2y^2 + 2\lambda y + 1\right)$

$\Rightarrow \lambda^2x^2y^2 - 2\lambda xy^2 + y^2 + \lambda^2x^2 - 2\lambda x + 1$

$= \lambda^2x^2y^2 + 2\lambda x^2y + x^2 + \lambda^2y^2 + 2\lambda y + 1$

$\Rightarrow \lambda^2\left(x^2 - y^2\right) - 2\lambda\left(xy^2 + x^2y + x + y\right) = 0$

$\Rightarrow \lambda^2(x+y)(x-y) - 2\lambda$

$[xy(x+y) + (x+y)] = 0$

$\Rightarrow \lambda(x+y)[\lambda(x-y) - 2xy - 2] = 0$

$\Rightarrow (x+y)[\lambda(x-y) - 2xy - 2] = 0$

$\Rightarrow \lambda(x-y) - 2xy - 2 = 0$

$\Rightarrow \frac{2xy+2}{x-y} = \lambda \Rightarrow \frac{xy+1}{x-y} = \frac{\lambda}{2}$

$\Rightarrow \frac{\left(x\frac{dy}{dx}+y\right)(x-y)-(xy+1)\left(1-\frac{dy}{dx}\right)}{(x-y)^2} = 1$

This is the first order differential equation and clearly degree of $\frac{dy}{dx}$ **is** 1**. Hence degree of the differential equation is 1.**

Q.3 Answer:

2

Solution:

$\frac{dx}{dy} + \int y \cdot dx = x^3 \Rightarrow \int y \cdot dx = x^3 - \frac{dx}{dy}$

$\Rightarrow 1 + \frac{dy}{dx}\left(\int y \cdot dx\right) = x^3 \cdot \frac{dy}{dx}$

Differentiate both sides w.e.t. $x \Rightarrow 0 + \frac{dy}{dx}(y) + \left(\int y \cdot dx\right)\left(\frac{d^2y}{dx^2}\right) = x^3 \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx}\left(2x^2\right)$

$\Rightarrow y \cdot \frac{dy}{dx} + \frac{d^2y}{dx^2}\left[x^3 - \frac{dx}{dy}\right] = x^3 \cdot \frac{d^2y}{dx^2} + 2x^2\frac{dy}{dx}$

$\Rightarrow y\frac{dy}{dx} + x^3\frac{d^2y}{dx^2} - \left(\frac{dx}{dy}\right)\left(\frac{d^2y}{dx^2}\right) = x^3\frac{d^2y}{dx^2} + 2x^2\frac{dy}{dx}$

$\Rightarrow y\frac{dy}{dx} - \frac{dx}{dy} \cdot \frac{d^2y}{dx^2} = 2x^2 \cdot \frac{dy}{dx}$

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Multiplying both side by $\frac{dy}{dx}$ $y\left(\frac{dy}{dx}\right)^2 - \frac{d^2y}{dx^2} = 2x^2\left(\frac{dy}{dx}\right)^2$

$\Rightarrow \frac{d^2y}{dx^2} + \left(2x^2 - y\right)\left(\frac{dy}{dx}\right)^2 = 0$

Order $= 2$**, degree** $= 1$

Q.4 Answer:

1,1

Solution:

The eq. of parabolas having vertex at $(0, 0)$ **\& focus at** $(a, 0)$**, where** $(a > 0)$ **is :**

$y^2 = 4ax \quad \ldots$

2y. $\frac{dy}{dx} = 4a$ **[on differentiating]**

On putting the value of $(4a)$ **in eq. (1) we get,**

$2x \cdot \frac{dy}{dx} - y = 0$

in order $= 1$ **\&** **degree** $= 1$

Q.5 Answer:

1,1

Solution:

Since order of the highest derivative in the given diff. equation is 1 and exponent of the derivative is also 1 therefore degree and order is $(1, 1)$

Q.6 Answer:

2

Solution:

Given differential equation is $y = x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^{-1}$

Multiply by $\frac{dy}{dx}; y\frac{dy}{dx} = x\left(\frac{dy}{dx}\right)^2 + 1$

Since power of highest order derivative is $2. \therefore$ **degree** $= 2$

Q.7 Answer:

Order two

Solution:

Given differential equation, $\frac{d^2y}{dx^2} + x \cdot \frac{dy}{dx} + \sin y + x^2 = 0$

The order of highest derivative $= 2$ **and degree** $= 1.$

Q.8 Answer:

2

Solution:

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In the given equation,

K. $\frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^3\right]^{3/2}$

Squaring both the sides, $\mathrm{K}^2 \cdot \left(\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dx}^2}\right)^2 = \left[1 + \left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)^3\right]^3$

Degree of a differential equation is the highest power of the highest derivative in equation when derivatives are expressed as polynomial. Here degree of differential equation is 2**.**

Q.9 Answer:

2

Solution:

Degree of a differential equation is the power to which the highest derivative is raised when it is expressed as polynomial of derivatives. Given equation is $\left(\frac{d^3y}{dx^3}\right)^{2/3} - 3\left(\frac{d^2y}{dx^2}\right) + 5\left(\frac{dy}{dx}\right) + 4 = 0$

$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^{2/3} = 3\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 4$

Cube on both side, $\left(\frac{d^3y}{dx^3}\right)^2 = \left[3\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 4\right]^3$

Hence, degree $= 2$

Q.10 Answer:

1,2

Solution:

The given differential equation is $y = x\frac{dy}{dx} + \frac{dx}{dy}$

Multiplying both the sides by $\frac{dy}{dx}$ **We get** $\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)\mathrm{y} = \mathrm{x}\left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)^2 + 1$

$\Rightarrow \quad x\left(\frac{dy}{dx}\right)^2 - y\left(\frac{dy}{dx}\right) + 1 = 0$

Hence, order and degree of differential equation are 1 and 2 .

Q.11 Answer:

$(2, 3/2)$

Solution:

Degree of differential equation is always a positive integer. $\left(2, \frac{3}{2}\right)$ **can not be the feasible.**

Q.12 Answer:

Both 1 and 2

Solution:

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Statement 1: The given differential equation is not a polynomial equation in its derivatives. Hence, its degree is not defined.

Statement 2 : The highest order derivative in the given polynomial is 2 .

Q.13 Answer:

order 1 , degree 3

Solution:

$y^2 = 2c(x + \sqrt{c})$ **... (1)**

$\therefore 2yy_1 = 2c$

Now putting $c = yy_1$ in (1) we get

$y^2 = 2 \cdot yy_1 \left(x + \sqrt{yy_1}\right)$

$\Rightarrow \left(y^2 - 2xyy_1\right)^2 = 4\left(yy_1\right)^3$

$\Rightarrow \left(y^2 - 2xyy_1\right)^2 = 4y^3y_1^3$

$\Rightarrow$ **order 1, degree 3**

Q.14 Answer:

second order and first degree

Solution:

$Ax^2 + By^2 = 1$

$2Ax + 2By\frac{dy}{dx} = 0 \ldots (i)$

$2A + 2B\left(\frac{dy}{dx}\right)^2 + 2By\frac{d^2y}{dx^2} = 0 \ldots (ii)$

$-\frac{2By'}{x} + 2B(y)^2 + 2By'' = 0$

$\frac{-yy'}{x} + (y)^2 + yy'' = 0$

Order $= 2$, Degree $= 1$

Q.15 Answer:

$1, 2$

Solution:

Equation $y^2 = 4a9x - h\}$

$2yy_1 = 4a \Rightarrow yy_1 = 2a$

$yy_2 - y_2^1 = 0$

Hence

(b) is the correct answer.

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